

1	The curve with equation $y = f(x)$, where $f(x)$ is a cubic polynomial, has a maximum point with coordinates $\left(-2, \frac{34}{3}\right)$ and a minimum point with coordinates $\left(3, -\frac{19}{2}\right)$. Find the equation of the curve. [4]
2	Referred to the origin O, the points A, B, P and Q have position vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{p}$ and $\mathbf{q}$ respectively, such that $\mathbf{a} = 2$, $\mathbf{b}$ is a unit vector, and the angle between $\mathbf{a}$ and $\mathbf{b}$ is $\frac{\pi}{4}$. (i) Give a geometrical interpretation of $\mathbf{b} \cdot \mathbf{a}$. [1] (ii) Find $\mathbf{a} \times \mathbf{b}$, leaving your answer in exact form. [2] It is also given that $\mathbf{p} = 3\mathbf{a} + (\mu + 2)\mathbf{b}$ and $\mathbf{q} = (\mu + 3)\mathbf{a} + \mu\mathbf{b}$, where $\mu \in \mathbb{R}$. (iii) Show that $\mathbf{p} \times \mathbf{q} = (\mu^2 + 2\mu + 6)(\mathbf{b} \times \mathbf{a})$. [3] (iv) Hence find the smallest area of the triangle OPQ as μ varies. [3]
3	The function f is defined by $f : x \mapsto \frac{1}{3} \tan\left(\frac{x}{3}\right) \text{ for } x \in \mathbb{R}, 0 \leq x < \frac{3\pi}{2}.$ (i) Sketch the graph of $y = f(x)$, indicating clearly the vertical asymptote. [2] (ii) State the equation of the line of reflection between the graphs of $y = f(x)$ and $y = f^{-1}(x)$, and hence sketch the graph of $y = f^{-1}(x)$ on the same diagram, indicating clearly the horizontal asymptote. [2] The solutions to the equation $f(x) = f^{-1}(x)$ are $x = 0$ and $x = \alpha$, where $0 < \alpha < \frac{3\pi}{2}$. (iii) Using the diagram drawn, find, in terms of α, the area of the region bounded by the curves $y = f(x)$ and $y = f^{-1}(x)$. [5] Another function g is defined by $g : x \mapsto e^x \text{ for } x \in \mathbb{R}, x \geq -2.$ (iv) Show that the composite function gf exists and define gf in a similar form. [3]
4	(a) The complex numbers z and w satisfy the simultaneous equations $z + w^* + 5i = 10 \quad \text{and} \quad w ^2 = z + 18 + i.$ Find z and w. [4]

	(b) (i) It is given that $2+i$ is a root of the equation $z^2 - 5z + 7 + i = 0$. Find the second root of the equation in cartesian form, showing your working clearly. [2] (ii) Hence find the roots of the equation $-iw^2 + 5w + 7i - 1 = 0$. [2] (c) The complex number z is given by $z = -a + ai$, where a is a positive real number. (i) It is given that $w = -\frac{\sqrt{2}z^*}{z^4}$. Express w in the form $re^{i\theta}$, in terms of a, where $r > 0$ and $-\pi < \theta \leq \pi$. [4] (ii) Find the two smallest positive whole number values of n such that $\text{Re}(w^n) = 0$. [3]																				
5	A planning committee of 12 students consisting of one male and one female student from each of the 6 Arts stream classes (Class A to Class F) in a junior college is to be formed for the Humanities Seminar. There are 10 male and 10 female students in Class A. (i) How many ways can the representatives from Class A be chosen? [1] The committee meets for their first planning meeting and is seated at a round table. (ii) How many ways can the committee be seated if all the members need to be seated together with the member from the same class? [2] At the seminar, the committee members are to be seated in a row of 14 seats in the theatre together with the Principal and the Guest of Honour. The chairperson and the secretary are selected from the committee and they are both from Class F. (iii) How many ways can this be done if the Principal and the Guest of Honour occupy the middle seats and the committee members are seated together with the member from the same class except for the chairperson and the secretary? [4]																				
6	The table below shows the petrol mileage, y km/L and the weight, x kg in thousands for various car models in the year 1995. <table><tr><td>x</td><td>3.5</td><td>3</td><td>2.75</td><td>2.5</td><td>2.25</td><td>2</td><td>1.75</td><td>1.5</td><td>1.25</td></tr><tr><td>y</td><td>7.5</td><td>8.0</td><td>8.5</td><td>8.7</td><td>10.0</td><td>k</td><td>13.5</td><td>16.8</td><td>18.0</td></tr></table> (i) The equation of the regression line of y on x is $y = 22.51355 - 4.908387x$. Show that $k = 11.0$. [2] (ii) Draw a scatter diagram to illustrate the data. [1]	x	3.5	3	2.75	2.5	2.25	2	1.75	1.5	1.25	y	7.5	8.0	8.5	8.7	10.0	k	13.5	16.8	18.0
x	3.5	3	2.75	2.5	2.25	2	1.75	1.5	1.25												
y	7.5	8.0	8.5	8.7	10.0	k	13.5	16.8	18.0												

	(iii) With reference to the scatter diagram and context of the question, explain why model (C) below is the most appropriate for modelling the data as compared to the other 2 models. (A) $y = a + bx$, where a is positive and b is negative, (B) $y = a + b \ln x$, where a is positive and b is negative, (C) $y = a + \frac{b}{x}$, where a and b are positive. [1] (iv) Calculate the least squares estimates of a and b for model (C). [1] (v) Predict the weight of the car if the petrol mileage is 12 km/L. Comment on the reliability of your prediction. [2] (vi) Suppose there was an error in recording the y values and all the y values must be increased by a constant M km/L, state any change you would expect in the values of (a) $\bar{y}$, [1] (b) standard deviation of y and [1] (c) the correlation coefficient. [1]
7	(a) The random variable X follows a binomial distribution $B(10, p)$. (i) Given that X has two modes, $X = 4$ and $X = 5$, find the exact value of p. [2] (ii) Given instead that $P(X \leq 9) = \frac{1023}{1024}$, find the exact value of p. [2] (b) The random variable Y follows a binomial distribution $B(500, 0.5)$. A sample of 30 independent values of Y is recorded. (i) Find the probability that all the values recorded are less than or equal to 256. [2] (ii) The mean of the 30 values is calculated. Estimate the probability that this sample mean is less than or equal to 256, stating clearly the approximation used. [3] (iii) Explain why the probability found in part (ii) is larger than that found in part (i). [1]
8	A trading card game has rectangular cards of nominal size 64 mm wide and 89 mm long. However, due to the limited precision of the machine used to cut the cards to size, the widths of the trading cards follow a normal distribution with mean 64 mm and standard deviation 0.3 mm. The lengths of the trading cards follow an independent normal distribution with mean 89 mm and standard deviation 0.45 mm. The perimeter of the trading cards is twice the sum of its length and width.

	(i) Trading cards with length 90 mm and above are called “tall” cards. Find the percentage of trading cards that are “tall”. [1] (ii) Write down the distribution of the perimeter of the trading cards, in mm, and find the perimeter that is exceeded by 8% of the trading cards. [4] A brand of rectangular card sleeves are manufactured for the trading cards and the widths of the card sleeves follow a normal distribution with mean 66 mm and standard deviation 0.45 mm, whereas the lengths of the card sleeves follow an independent normal distribution with mean 91 mm and standard deviation 0.675 mm. For a card sleeve to fit the trading card nicely, the dimensions of the sleeves must be larger than the dimensions of the trading card, but there should only be a maximum allowance of 1.2 mm on each side. (iii) Find the probability that a randomly chosen card sleeve fits a randomly chosen trading card nicely, stating clearly the parameters of any distribution used. [5]
9	A computer hard drive manufacturer claims that the mean usage hours before failure of their R series hard drives is 50 thousand hours. A technology columnist wishes to investigate this claim and collected the usage hours, t thousand hours for each of the 50 randomly chosen hard drives which were submitted to the local service centre for drive failures. The data is summarized as follows. $n = 50 \qquad \Sigma t = 2384.5 \qquad \Sigma t^2 = 115885.23$ The technology columnist wants to use hypothesis testing to test whether the mean usage hours before failure of a hard drive is different from what the manufacturer has stated. (i) Explain whether it is necessary for the columnist to know about the distribution of the usage hours before failure of the drives in order to carry out a hypothesis test. [1] (ii) Find the unbiased estimates of the population mean and variance and carry out the test at 1% level of significance for the columnist. [6] The columnist published the data and the results of the hypothesis testing in an online article. (iii) Suggest a reason why the test result might not be useful to a reader of the article who is deciding whether to buy an R series hard drive from the manufacturer. [1] (iv) State an alternative hypothesis that is more relevant to the decision making process and explain whether the result will differ from the earlier test carried out by the columnist at 1% level of significance. [2] (v) State a necessary assumption that was made for all the tests carried out. [1]

10 A box contains 2 red balls, 3 green balls and x blue balls, where $x \in \mathbb{Z}, x \geq 5$. A game is played where the contestant picks 5 balls from the box without replacement. The total score, S , for the contestant is the sum of the number of green balls chosen and thrice the number of red balls chosen. The blue balls will not contribute any points, unless all 5 balls are blue. If all the 5 balls are blue, the score will be 25 points.

(i) Show that $P(S = 6) = \frac{20x(x^2 - 3x + 14)}{(x+5)(x+4)(x+3)(x+2)(x+1)}$. [2]

(ii) Given that $P(S = 6) = \frac{5}{63}$, calculate x . [2]

(iii) Complete the probability distribution table for S . [4]

s	1	2	3	4	5	6	7	8	9	25
$P(S = s)$		$\frac{5}{42}$	$\frac{5}{63}$		$\frac{5}{21}$	$\frac{5}{63}$		$\frac{5}{84}$	$\frac{1}{252}$	

(iv) Evaluate $E(S)$ and find the probability that S is more than $E(S)$. [2]

(v) Find the probability that there are no green balls drawn given that S is more than $E(S)$. [2]